

Nassau Candy Distributor: Regional Reallocation & Shipping Optimization

Research Paper / Project Report

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Program: Unified Mentor - Data Analyst Internship Project

Project Type: Data Analytics, Business Intelligence and Machine Learning

This report documents the analytical workflow, exploratory analysis, historical regional reallocation scenario, lead-time prediction model, dashboard implementation, key findings, limitations, and deployment.

GitHub: <https://github.com/Praveshyadav47/Nassau-Candy-Factory-Optimization>

Live Dashboard: <https://yluxmz8tgem6zpaxvq2dc9.streamlit.app/>

Abstract

This project analyzes the Nassau Candy Distributor dataset to understand sales performance, profitability, regional differences, shipping lead times, and product-level operational patterns. The analysis combines exploratory data analysis with historical scenario analysis and a supervised machine-learning model for shipping lead-time prediction. An interactive Streamlit dashboard was developed so that users can filter the data by region, division, and ship mode, inspect product-region performance, evaluate a historical regional reallocation scenario, and generate lead-time predictions.

The completed dashboard reports 8,549 unique Order IDs, total sales of \$141,783.63, gross profit of \$93,442.80, an average lead time of approximately 1,320.8 days, and an average profit margin of 66.51%. The lead-time prediction model is a Random Forest regression model with test-set MAE of 179.75 days, RMSE of 203.05 days, and R2 of 0.4170. The regional reallocation component is explicitly treated as historical scenario analysis rather than a guarantee of future operational improvement.

Keywords: Data Analytics, Supply Chain Analytics, Lead Time, Regional Analysis, Random Forest, Streamlit, Python, Business Intelligence

1. Introduction

Distribution businesses need visibility into where sales are generated, how profitable products are, how shipping performance varies across operational dimensions, and where historical performance suggests potential opportunities for improvement. The Nassau Candy Distributor project addresses these questions through a reproducible Python-based analytics workflow and an interactive dashboard.

The project is designed as an end-to-end data analyst internship deliverable: data preparation, exploratory analysis, business interpretation, machine learning, dashboard development, GitHub version control, and cloud deployment.

2. Problem Statement

The central problem is to convert transaction-level distributor data into actionable information about regional performance, profitability, shipping lead times, and potential operational allocation scenarios. A secondary objective is to determine whether historical order characteristics provide useful signal for predicting shipping lead time.

3. Objectives

- Clean and prepare the distributor transaction data for analysis.
- Calculate and analyze shipping lead time from order and ship dates.
- Measure sales, gross profit, and profit margin across regions and products.
- Compare lead-time performance across shipping modes and regional/product combinations.
- Build a historical regional reallocation scenario using Product x Region performance.
- Train and evaluate a machine-learning model for lead-time prediction.
- Develop an interactive Streamlit dashboard for business users.
- Publish the project through GitHub and Streamlit Community Cloud.

4. Dataset and Data Preparation

The project uses the supplied Nassau Candy Distributor CSV dataset. The completed analysis contains 10,194 transaction-level records used for profit-margin descriptive statistics. Key fields used in the dashboard include Order ID, Order Date, Ship Date, Sales, Gross Profit, Cost, Units, Product Name, Ship Mode, Division, and Region.

Analytical field	Purpose
Order ID	Unique-order counting and transaction identification
Order Date / Ship Date	Calculation of shipping lead time
Sales	Revenue analysis
Gross Profit / Cost	Profitability analysis
Units	Volume-related analysis and model input
Product Name	Product-level analysis
Ship Mode	Shipping-method comparison
Division	Operational/business segmentation
Region	Regional performance and allocation dimension

4.1 Derived Variables

- **Lead Time:** Ship Date - Order Date, measured in days.
- **Profit Margin %:** (Gross Profit / Sales) x 100.
- **Product x Region metrics:** order count, sales, gross profit, average lead time, and average profit margin.
- **Regional reallocation metrics:** current average lead time, best historical regional lead time, potential reduction, and reduction percentage.

4.2 Data Quality Observation

The resulting lead-time values are unusually large in magnitude for a conventional shipping KPI, with the dashboard reporting an overall mean of approximately 1,320.8 days. This is retained as a data-quality observation rather than silently removing observations. Before using the model or scenario results for real operational decisions, the underlying order and ship dates should be validated with the source system.

5. Exploratory Data Analysis

The exploratory analysis examined profitability, regional sales, shipping performance, product behavior, and interactions between product and region. The dashboard provides filters for Region, Division, and Ship Mode so that the same KPIs can be viewed for different operational slices.

KPI	Completed result
Unique Order IDs	8,549
Total Sales	\$141,783.63
Gross Profit	\$93,442.80
Average Lead Time	1,320.8 days
Average Profit Margin	66.51%
Profit Margin range	7.69% to 80.00%
Profit Margin median	66.67%

5.1 Profitability

The average profit margin is 66.51%, with a median of 66.67%. The observed minimum and maximum are 7.69% and 80.00%, respectively. The distribution indicates that profitability varies across transactions and should be considered

alongside product, regional, and sales-volume patterns rather than using a single margin figure as a complete performance measure.

5.2 Regional Performance

Sales and average lead time were summarized by the four observed regions: Atlantic, Gulf, Interior, and Pacific. The dashboard allows these regional metrics to be filtered further by division and ship mode, supporting more granular investigation of operational differences.

5.3 Product-Level Analysis

Product-region analysis was created to identify combinations with different sales, gross profit, profit margin, and lead-time profiles. This table is also the foundation for the regional reallocation simulator.

6. Historical Regional Reallocation Scenario

The regional reallocation component compares historical product performance across regions. For a selected product, the current average lead time is compared with the lowest observed historical average lead time among the available regions. The difference is presented as a potential reduction under a historical scenario.

The supplied dataset does not contain an explicit factory identifier. Therefore, Region is used as the operational allocation dimension. This is a modeling assumption and should not be interpreted as proof that physical factory relocation is possible.

6.1 Example: Wonka Bar - Milk Chocolate

Metric	Value
Current average lead time	1,314.26 days
Suggested historical region	Gulf
Best historical lead time	1,289.73 days
Potential reduction	24.53 days
Potential reduction	1.87%
Scenario confidence score	100%*

*The confidence score is a project-defined sample-size heuristic, not a statistical probability. The scenario is based on historical averages and does not guarantee future performance.

7. Machine Learning Methodology

A Random Forest Regressor was trained to estimate shipping lead time from order and operational attributes. Categorical variables such as Product Name, Ship Mode, Division, and Region are encoded for model use, while numeric fields include Sales, Units, Gross Profit, Cost, Order Year, and Order Month.

7.1 Model Evaluation

Metric	Test-set result	Interpretation
MAE	179.75 days	Average absolute prediction error
RMSE	203.05 days	Penalizes larger prediction errors more strongly
R2	0.4170	Explains about 41.7% of test-set variance under this evaluation

The model provides measurable predictive signal, but the error magnitude is large relative to the operational meaning of lead time. Predictions should therefore be used as analytical estimates rather than guaranteed delivery times. The underlying data quality should be validated before using the model in production decision-making.

8. Streamlit Dashboard

The project was converted into an interactive Streamlit application. The dashboard contains a title and methodology note, sidebar filters, KPI cards, regional charts, product-region analysis, a regional reallocation simulator, a lead-time prediction interface, model performance metrics, and a methodology/limitations section.

- Dashboard filters: Region, Division, and Ship Mode.
- KPI cards: unique orders, sales, gross profit, average lead time, and average profit margin.
- Business Overview: sales by region and average lead time by region.
- Product Analysis: product-region performance table.
- Regional Reallocation Simulator: historical scenario metrics for the selected product.
- Lead-Time Prediction: interactive Random Forest prediction form.
- Model Performance: MAE, RMSE, and R2.

9. Key Findings

- The project dataset contains 10,194 transaction records used for descriptive profitability analysis.
- There are 8,549 unique Order IDs represented in the dashboard.
- Total sales are \$141,783.63 and total gross profit is \$93,442.80.
- Average profit margin is 66.51%, with a median of 66.67%.
- Lead-time values are unusually high in magnitude and should be validated before operational use.
- Historical Product x Region comparisons can identify regions with lower observed average lead times for individual products.
- The Random Forest model achieved MAE of 179.75 days, RMSE of 203.05 days, and R2 of 0.4170.
- The dashboard turns the analysis into an interactive tool that can be explored without opening the Jupyter notebook.

10. Business Recommendations

- Validate Order Date and Ship Date definitions with the source system before using lead time as a production KPI.
- Investigate product-region combinations with persistently different lead times and assess whether the difference is operationally actionable.
- Use sales, gross profit, and lead time together when evaluating regional allocation rather than optimizing for a single KPI.
- Treat the regional reallocation simulator as a scenario-planning tool and validate recommendations with additional operational constraints.
- Collect or integrate explicit factory, warehouse, carrier, or fulfillment-center identifiers in future versions.
- Use the lead-time model as a decision-support estimate and retrain it when cleaner operational data becomes available.
- Continue monitoring model performance over time rather than treating the current test-set metrics as permanent.

11. Limitations and Future Scope

- The dataset does not contain an explicit factory identifier; Region is therefore used as the allocation dimension.
- Historical regional averages do not establish causality and do not guarantee future performance.
- The lead-time metric is unusually large, creating a data-quality concern that should be investigated.
- The Random Forest model's error remains substantial, so it should not be treated as a precise delivery-time forecasting system.
- Future work can add warehouse/factory identifiers, carrier information, distance, inventory availability, order priority, seasonality, and actual fulfillment constraints.
- Future modeling can compare additional regressors, cross-validation strategies, feature engineering, and error analysis by region and product.

12. Conclusion

The Nassau Candy Distributor project demonstrates an end-to-end data analytics workflow that connects transaction-level analysis with business interpretation, machine learning, and deployment. The analysis provides a consolidated view of sales, profitability, regional performance, product behavior, and shipping lead time. The regional reallocation simulator shows how historical Product x Region performance can be converted into scenario analysis, while the Random Forest model provides a baseline for lead-time prediction.

The project also highlights an important analytics principle: model and scenario outputs are only as reliable as the underlying data definitions. In particular, the unusually large lead-time values and absence of a factory identifier should be addressed before using these outputs for operational decisions. Within those limitations, the Streamlit dashboard provides a practical interface for exploring the completed analysis.

13. Project Deliverables

Deliverable	Status / Link
Jupyter analysis	Completed and included in GitHub repository
Streamlit dashboard	Deployed and live
GitHub repository	https://github.com/Praveshyadav47/Nassau-Candy-Factory-Optimization
Live dashboard	https://yluxmz8tgem6zpaxvq2dc9.streamlit.app/
Research paper	This report

14. References

- Unified Mentor - Project requirements and internship project brief for the Nassau Candy Distributor analysis.
- Project dataset: Nassau Candy Distributor.csv.
- Project Jupyter analysis and Streamlit implementation.
- Scikit-learn documentation and Random Forest regression methodology.
- Streamlit Community Cloud documentation for application deployment.

Declaration: The quantitative results in this report are based on the completed project analysis and dashboard outputs. Scenario results are historical comparisons and are not presented as guaranteed operational outcomes.